

## 미적분 Black Half Test 07. 정답 및 해설

$$\begin{aligned}
 1) \int_1^3 \frac{1}{\sqrt{(x-1)(3-x)}} dx &= \int_1^3 \frac{1}{\sqrt{1-(x-2)^2}} dx \\
 &= [\sin^{-1}(x-2)]_1^3 \\
 &= \sin^{-1}1 - \sin^{-1}(-1) \\
 &= \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi
 \end{aligned}$$

정답 : ③

$$2) \int_{-\infty}^{\infty} (1+x^2)e^{-x^2} dx = \int_{-\infty}^{\infty} e^{-x^2} dx + \int_{-\infty}^{\infty} x^2 e^{-x^2} dx \text{ 이고}$$

$$(i) \int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \text{ 이고}$$

$$f(x) = e^{-x^2} \text{은 우함수이므로 } \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \text{이다.}$$

$$(ii) \int_0^{\infty} x^2 e^{-x^2} dx = \frac{\sqrt{\pi}}{4} \text{ 이고}$$

$$f(x) = x^2 e^{-x^2} \text{은 우함수이므로 } \int_{-\infty}^{\infty} x^2 e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \text{이다.}$$

$$(i), (ii) \text{에 의하여 } \int_{-\infty}^{\infty} (1+x^2)e^{-x^2} dx = \sqrt{\pi} + \frac{\sqrt{\pi}}{2} = \frac{3}{2}\sqrt{\pi} \text{이다.}$$

정답 : ④

$$3) x = \frac{2}{t} \text{라 치환하면 } dx = -\frac{2}{t^2} dt \text{ 이고}$$

$$\begin{aligned}
 &\int_0^{\infty} \frac{\ln x}{x^2+4} dx \\
 &= \int_{\infty}^0 \frac{\ln\left(\frac{2}{t}\right) - 2}{\frac{4}{t^2} + 4} \frac{-2}{t^2} dt \\
 &= \frac{1}{2} \int_0^{\infty} \frac{\ln\left(\frac{2}{t}\right)}{t^2+1} dt \\
 &= \frac{1}{2} \int_0^{\infty} \frac{\ln 2 - \ln t}{t^2+1} dt \\
 &= \frac{1}{2} \int_0^{\infty} \frac{\ln 2}{t^2+1} dt - \frac{1}{2} \int_0^{\infty} \frac{\ln t}{t^2+1} dt \\
 &= \frac{1}{2} \int_0^{\infty} \frac{\ln 2}{t^2+1} dt \quad \left(\because \int_0^{\infty} \frac{\ln x}{x^2+1} dx = 0\right) \\
 &= \frac{\ln 2}{2} [\tan^{-1}t]_0^{\infty} \\
 &= \frac{\pi \ln 2}{4}
 \end{aligned}$$

$$4) \ln x = t \text{로 치환하면 } \frac{1}{x} dx = dt \text{이며}$$

$$\begin{aligned}
 \int_1^2 \frac{1}{x(\ln x)^p} dx &= \int_0^{\ln 2} \frac{1}{t^p} dt \text{이며} \\
 \text{수렴 조건은 } p < 1 \text{이다. (단, } p > 0) \\
 \therefore 0 < p < 1
 \end{aligned}$$

$$\begin{aligned}
 5) I &= \int_0^{\infty} \left( \frac{1}{\sqrt{x^2+9}} - \frac{a}{x+1} \right) dx \\
 &= \left[ \sinh^{-1}\left(\frac{x}{3}\right) - a \ln(x+1) \right]_0^{\infty}
 \end{aligned}$$

$$\begin{aligned}
 &= \left[ \ln\left(\frac{x}{3} + \sqrt{\frac{x^2}{9} + 1}\right) - a \ln(x+1) \right]_0^{\infty} \\
 &= \left[ \ln\left(\frac{x + \sqrt{x^2+9}}{3(x+1)^a}\right) \right]_0^{\infty} \text{이다. 따라서 } a=1 \text{일 때, 수렴하며} \\
 &= \left[ \ln\left(\frac{x + \sqrt{x^2+9}}{3(x+1)}\right) \right]_0^{\infty} \\
 &= \ln\left(\frac{2}{3}\right) - \ln\left(\frac{3}{3}\right) = \ln\left(\frac{2}{3}\right)
 \end{aligned}$$

이다. 그러므로  $a=1, b=1, c=2, d=3$ 이고  
 $a+b+c-d=1$ 이다.

정답 : ①

$$6) f(x) \text{는 확률밀도함수이므로}$$

$$\begin{aligned}
 \int_{-\infty}^{\infty} f(x) dx &= c \int_0^1 \tan^{-1} x dx = c \left[ x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]_0^1 \\
 &= c \left( \frac{\pi}{4} - \frac{1}{2} \ln 2 \right) = 1
 \end{aligned}$$

$$\text{이므로 } c = \frac{1}{\frac{\pi}{4} - \frac{1}{2} \ln 2} = \frac{4}{\pi - 2 \ln 2} \text{이다.}$$

$$\begin{aligned}
 \text{따라서 평균 } E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \frac{4}{\pi - 2 \ln 2} \int_0^1 x \tan^{-1} x dx \\
 &= \frac{4}{\pi - 2 \ln 2} \left[ \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} (x - \tan^{-1} x) \right]_0^1 \\
 &= \frac{4}{\pi - 2 \ln 2} \left( \frac{\pi}{4} - \frac{1}{2} \right) = \frac{\pi - 2}{\pi - 2 \ln 2} \text{이다.}
 \end{aligned}$$

정답 : ②

$$\begin{aligned}
 7) \int_0^{\infty} \frac{e^{-x^2}}{|x-2|^{3/2}} dx &= \int_0^3 \frac{e^{-x^2}}{|x-2|^{3/2}} dx + \int_3^{\infty} \frac{e^{-x^2}}{|x-2|^{3/2}} dx \\
 &\approx \int_0^3 \frac{1}{|x-2|^{3/2}} dx + \int_3^{\infty} \frac{1}{e^{x^2}} dx \\
 &= \text{발산} + \text{수렴} = \text{발산}
 \end{aligned}$$

$$\therefore \int_0^{\infty} \frac{1}{1+x^4} dx = \text{수렴}$$

$$\text{ㄷ. } \sqrt{\ln x} = t \text{라고 치환하면 } \ln x = t^2, e^{t^2} = x \text{이므로 } 2te^{t^2} dt = dx \text{이다.}$$

$$\begin{aligned}
 \int_{2026}^{\infty} e^{-\sqrt{(\ln x)^3}} dx &= \int_{\sqrt{\ln 2026}}^{\infty} e^{-t^3} 2te^{t^2} dt \\
 &\approx \int_{\sqrt{\ln 2026}}^{\infty} \frac{1}{e^{t^3-t^2}} dt \approx \int_{\sqrt{\ln 2026}}^{\infty} \frac{1}{e^{t^3}} dt = \text{수렴}
 \end{aligned}$$

$$\text{ㄹ. } \int_0^{2026} \frac{\cos x}{\sqrt{x}} dx \approx \int_0^{2026} \frac{1}{\sqrt{x}} dx = \text{수렴}$$

정답 : ③

$$8) \int_0^{\pi} x f(\sin x) dx = \int_0^{\pi} \frac{\pi}{2} f(\sin x) dx \text{을 이용하자.}$$

$$\begin{aligned}
 &\int_0^1 x \ln(\sin(\pi x)) dx \\
 &= \frac{1}{\pi^2} \int_0^{\pi} t \ln(\sin(t)) dt \quad (\because \pi x = t) \\
 &= \frac{1}{\pi^2} \int_0^{\pi} \frac{\pi}{2} \ln(\sin(t)) dt \\
 &= \frac{1}{2\pi} \int_0^{\pi} \ln(\sin(t)) dt
 \end{aligned}$$

정답 : ②

$$= \frac{1}{2} \int_0^1 \ln(\sin(\pi x)) dx \quad (\because t = \pi x)$$

$$= \int_0^{\frac{1}{2}} \ln(\sin(\pi x)) dx \quad (\because \text{대칭성})$$

$$= -\frac{\ln 2}{2}$$

$$\text{참고. } \int_0^{\frac{1}{2}} \ln(\sin \pi x) dx = \int_0^{\frac{1}{2}} \ln(\cos \pi x) dx = -\frac{\ln 2}{2}$$

9)

$$\int_0^1 x^2 \left( \ln \frac{1}{x} \right)^{\frac{1}{2}} dx$$

$$= \int_0^1 x^2 (-\ln x)^{\frac{1}{2}} dx \quad (\because -\ln x = t \text{로 치환})$$

$$= \int_0^\infty e^{-3t} t^{\frac{1}{2}} dt \quad (\because 3t = w \text{로 치환})$$

$$= \int_0^\infty e^{-w} \left( \frac{w}{3} \right)^{\frac{1}{2}} \frac{1}{3} dw = \frac{1}{3\sqrt{3}} \int_0^\infty \sqrt{w} e^{-w} dw$$

$$= \frac{1}{3\sqrt{3}} \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{3}\pi}{18}$$

정답 : ④

10)

$$\int_1^\infty \frac{\ln(x-1)}{x^2-2x+2} dx$$

$$= \int_1^\infty \frac{\ln(x-1)}{(x-1)^2+1} dx$$

$$= \int_0^\infty \frac{\ln t}{t^2+1} dt$$

$$= 0 \quad (\text{암기})$$

11)

$$\int_{1+\sqrt{3}}^\infty \frac{1}{x^3+8} dx$$

$$= \int_{1+\sqrt{3}}^\infty \frac{1}{(x+2)(x^2-2x+4)} dx$$

$$= \int_{1+\sqrt{3}}^\infty \frac{\frac{1}{12}}{x+2} + \frac{-\frac{1}{12}x + \frac{1}{3}}{x^2-2x+4} dx$$

$$= \int_{1+\sqrt{3}}^\infty \frac{\frac{1}{12}}{x+2} + \frac{-\frac{1}{24}(2x-2) + \frac{1}{4}}{x^2-2x+4} dx$$

$$= \left[ \frac{1}{12} \ln(x+2) - \frac{1}{24} \ln(x^2-2x+4) + \frac{1}{4\sqrt{3}} \tan^{-1} \left( \frac{x-1}{\sqrt{3}} \right) \right]_{1+\sqrt{3}}^\infty$$

$$= \left[ \frac{1}{12} \ln \left( \frac{x+2}{\sqrt{x^2-2x+4}} \right) + \frac{1}{4\sqrt{3}} \tan^{-1} \left( \frac{x-1}{\sqrt{3}} \right) \right]_{1+\sqrt{3}}^\infty$$

$$= \frac{1}{4\sqrt{3}} \times \frac{\pi}{2} - \frac{1}{12} \ln \left( \frac{3+\sqrt{3}}{\sqrt{6}} \right) - \frac{1}{4\sqrt{3}} \times \frac{\pi}{4}$$

$$= \frac{\sqrt{3}\pi}{48} - \frac{1}{12} \ln \left( \frac{3+\sqrt{3}}{\sqrt{6}} \right)$$

$$\therefore a \times b = \frac{1}{4}$$

정답 : ④

12)

라이프니츠 법칙에 의해

$$f'(x) = \int_0^\infty -\frac{t^3}{2\sqrt{xt}} e^{-\sqrt{xt}} dt \quad \text{에서}$$

$$f'(1) = -\frac{1}{2} \int_0^\infty t e^{-\sqrt{t}} dt$$

$$= -\int_0^\infty \alpha^3 e^{-\alpha} d\alpha \quad (\because \sqrt{t} = \alpha \text{로 치환})$$

$$= -\Gamma(4) = -3! = -6$$

정답 : ③

13)

$\ln x = t$ 로 치환하면  $x = e^t$ 이고  $dx = e^t dt$ 이다.

$$a_n = \int_1^\infty \frac{(\ln x)^{n-1}}{x^3} dx$$

$$= \int_0^\infty t^{n-1} e^{-3t} e^t dt$$

$$= \int_0^\infty t^{n-1} e^{-2t} dt \quad (2t = u)$$

$$= \frac{1}{2} \int_0^\infty \left( \frac{u}{2} \right)^{n-1} e^{-u} du$$

$$= \frac{1}{2^n} \int_0^\infty e^{-u} u^{n-1} du$$

$$= \frac{1}{2^n} \Gamma(n) \quad (n : \text{자연수})$$

$$= \frac{(n-1)!}{2^n}$$

$$\sum_{n=1}^\infty \frac{1}{a_n} = \sum_{n=1}^\infty \frac{2^n}{(n-1)!}$$

$$= \sum_{n=0}^\infty \frac{2^{n+1}}{n!}$$

$$= 2 \sum_{n=0}^\infty \frac{2^n}{n!} = 2e^2 \quad (\because e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots)$$

정답 : ③

14)

$$\lim_{n \rightarrow \infty} \left( \int_{-1}^{-\frac{1}{n}} \frac{1}{x} dx + \int_{\frac{1}{n}}^1 \frac{1}{x} dx \right)$$

$$= \lim_{n \rightarrow \infty} (0) = 0 \quad (\because \text{대칭성}) \quad [\text{참}]$$

ㄴ. 이상적분의 정의에 의하여  $\int_{-1}^1 \frac{1}{x} dx$ 은 발산한다. [거짓]

ㄷ.  $f : (0, \infty) \rightarrow \mathbb{R}$ 가 증가함수이고 이상적분

$$\int_0^\infty f(x) dx \text{가 수렴하면 } \lim_{x \rightarrow \infty} f(x) = 0 \text{이다.} \quad [\text{참}]$$

ㄹ. 함수  $f : (0, \infty) \rightarrow \mathbb{R}$ 에 대한 이상적분

$$\int_0^\infty |f(x)| dx \text{가 수렴하면 } \lim_{x \rightarrow \infty} f(x) = 0 \text{이다.} \quad [\text{거짓}]$$

정답 : ②

15)

$$\int_{-\infty}^\infty \frac{x^2}{x^4 - x^2 + 1} dx$$

$$= \int_0^\infty \frac{2}{x^2 - 1 + \frac{1}{x^2}} dx$$

$$= \int_0^\infty \frac{1 - \frac{1}{x^2}}{x^2 - 1 + \frac{1}{x^2}} + \frac{1 + \frac{1}{x^2}}{x^2 - 1 + \frac{1}{x^2}} dx$$

$$= \int_0^\infty \frac{1 - \frac{1}{x^2}}{\left(x + \frac{1}{x}\right)^2 - 3} + \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 1} dx$$

정답 : ②

$$= \frac{1}{2\sqrt{3}} \left[ \ln \left( \frac{x + \frac{1}{x} - \sqrt{3}}{x + \frac{1}{x} + \sqrt{3}} \right) \right]_{0+}^{\infty} + \left[ \tan^{-1} \left( x - \frac{1}{x} \right) \right]_{0+}^{\infty}$$

$$= \pi$$

정답 : ②



장황수학